🧠 Scenario-Based MCQs: Microsoft Fabric (Enterprise Level)

🏢 Scenario 1: Unified Data Platform Migration

A large enterprise is migrating from multiple legacy systems (SQL Server, Data Lake, Power BI Service) into a unified platform using Microsoft Fabric.

Q1

The team wants a **single storage layer** for structured and unstructured data with ACID transactions.

What should they use?

A. Data Warehouse
B. Lakehouse
C. Power BI Dataset
D. Dataflow Gen2

Q2

Data engineers need to process large-scale data using Spark while maintaining SQL accessibility.

What is the BEST choice?

A. Data Warehouse
B. Lakehouse with Delta tables
C. Power BI
D. Data Activator

Q3

Business users want to build reports directly on top of the unified data without duplication.

What should be implemented?

A. Export data to Excel
B. DirectLake mode
C. Import mode only
D. CSV extracts

🔄 Scenario 2: Data Ingestion & Pipeline Orchestration

A company ingests data from APIs, databases, and files into Fabric daily.

Q4

They need a **low-code ingestion tool** for transforming and loading data.

What should they use?

A. Notebook
B. Dataflow Gen2
C. SQL endpoint
D. Power BI Dashboard

Q5

The pipeline must orchestrate multiple steps including ingestion, transformation, and model refresh.

What is the BEST solution?

A. Manual execution
B. Fabric Data Pipelines
C. Excel macros
D. Power BI visuals

Q6

A pipeline fails intermittently due to API latency.

What is the BEST enterprise approach?

A. Ignore failures
B. Implement retry policies and logging
C. Reduce data size
D. Run pipeline once

🧱 Scenario 3: Lakehouse Architecture & Delta Tables

Q7

A team wants version control and time travel capabilities on their data.

What feature should they use?

A. CSV files
B. Delta Lake tables
C. Excel storage
D. JSON files

Q8

Data updates need to support **merge (insert/update/delete)** operations efficiently.

What is the BEST approach?

A. Append-only data
B. Delta Lake MERGE
C. Recreate tables daily
D. Use Excel

Q9

Which layer in Medallion architecture stores raw ingested data?

A. Gold
B. Silver
C. Bronze
D. Semantic

📊 Scenario 4: Data Warehousing & SQL Analytics

Q10

A finance team requires **high-performance SQL queries** on structured data.

What should be used?

A. Lakehouse only
B. Fabric Data Warehouse
C. Power BI only
D. Dataflow

Q11

The team wants separation of compute and storage for scalability.

Which Fabric capability supports this?

A. Dataflows
B. Warehouse engine
C. Excel
D. Dashboards

Q12

Complex joins and aggregations are slow in Lakehouse SQL endpoint.

What is the BEST solution?

A. Use CSV
B. Move workload to Data Warehouse
C. Reduce joins
D. Use Power BI

📈 Scenario 5: Power BI Integration

Q13

A report needs real-time data access without scheduled refresh.

What mode should be used?

A. Import
B. DirectLake
C. Excel export
D. Snapshot

Q14

Users complain about slow dashboards due to large datasets.

What is the BEST optimization?

A. Increase visuals
B. Use aggregations and semantic model optimization
C. Add more columns
D. Export data

Q15

A single version of truth is required across reports.

What should be implemented?

A. Multiple datasets
B. Centralized semantic model
C. Excel sheets
D. CSV sharing

🔐 Scenario 6: Governance & Security

Q16

Sensitive financial data must be restricted based on user roles.

What should be used?

A. Dataflows
B. Row-Level Security (RLS)
C. Pipelines
D. Notebooks

Q17

The organization wants to track data lineage and dependencies.

Which feature helps?

A. Dashboards
B. Lineage view in Fabric
C. Excel logs
D. Manual tracking

Q18

Data access must comply with enterprise security policies across all workloads.

What is the BEST approach?

A. Manual permissions
B. Centralized governance in Fabric workspace
C. Share links
D. Export data

⚙️ Scenario 7: Real-Time Analytics & Monitoring

Q19

A company wants to trigger alerts when sales drop below threshold in real-time.

What should they use?

A. Power BI import mode
B. Data Activator
C. Excel formulas
D. Dataflows

Q20

Streaming data from IoT devices needs to be processed and visualized instantly.

What is the BEST solution?

A. Batch processing
B. Real-Time Analytics in Fabric
C. CSV storage
D. Manual refresh